David Amebley

Ph.D. Student in Computer Science, The University of Alabama

davidamebley.github.io | github.com/davidamebley | linkedin.com/in/amebley

Education

Ph.D. in Computer Science 2025–present
The University of Alabama
- Advisor: Dr. Tu Le

M.Sc. in Computer Science 2020–2022
University of Eastern Finland

B.Sc. in Information Technology 2014–2018
Garden City University College, Ghana

Research Experience

Graduate Researcher 2026–present
Human-Centered Computing Lab, The University of Alabama
- Human-centered research on developer decisions and usability barriers in post-quantum cryptography adoption (ongoing).

Graduate Researcher 2025–2026
Trustworthy AI Lab, The University of Alabama
- Privacy evaluation and membership inference studies of vision-language models.

Master's research project 2022
University of Eastern Finland
- Augmented reality prototype for microsurgical skills training.

Teaching Experience

Graduate Teaching Assistant Spring and Fall 2026
The University of Alabama
- CS 200: Software Design and Engineering. Assist with course instruction, lead lab sessions, hold office hours, grade coursework, and support students with course concepts.

Teaching Assistant Fall 2025
The University of Alabama
- CS 470/570: Computer Algorithms. Grading and instructional support.

IT Teacher 2011–2020
Shining Star International School, Kumasi, Ghana
- Taught computing and digital literacy, organizing coding competitions and project-based learning activities.
- Mentored students in foundational programming and problem-solving skills.

Industry Experience

Full-Stack Developer 2023–2025
Hack Junction, Finland

- Developed and maintained Node.js backend services, debugging production issues and improving database schemas and reliability.
- Built React frontend features and implemented PostHog analytics to support usability improvements and data-informed product decisions.

Junior Full-Stack Developer 2022–2023
Integrify Academy, Finland

- Developed full-stack web applications using TypeScript, React, Node.js, and SQL, with AWS-related deployment workflows.
- Built CI/CD pipelines using Docker and GitHub Actions.

Publications

Conference papers

David Amebley and Sayanton Dibbo. "Are Neuro-Inspired Multi-Modal Vision-Language Models Resilient to Membership Inference Privacy Leakage?" *20th USENIX WOOT Conference on Offensive Technologies (WOOT 2026)*, pp. 263–285. Paper PDF

Raj Patel, Shaswata Mitra, **David Amebley**, Taye Akinrele, Sayanton Dibbo, Sudip Mittal, and Shahram Rahimi. "The Value of Spike Timing: A Leakage-Resistant Benchmark of SNN Design Choices for Network Intrusion Detection." Accepted at *IEEE CogMI 2026*. arXiv preprint

Journal manuscripts

Raj Patel, **David Amebley**, Taye Akinrele, Shaswata Mitra, Sayanton Dibbo, and Shahram Rahimi. "Tree-to-Spike: Tree-Guided Temporal Encoding for Compact Spiking Intrusion Detection." Submitted to *IEEE Transactions on Neural Networks and Learning Systems*.

Thesis

David Amebley. "Augmented Reality for Micro-Surgical Skill Training." Master's thesis, University of Eastern Finland, 2022.

Presentations

VLMLeaks: Membership Inference Attacks against Vision-Language Models October 2025
Poster presentation, Alabama Higher Education AI Exchange

Honors and Awards

Best Graduating Student 2018
Garden City University College

IMPIT Tuition Scholarship 2020–2022
University of Eastern Finland

Certifications and Training

AWS Certified Cloud Practitioner Earned 2023
Amazon Web Services

Responsible Conduct of Research and Human Subjects Research training 2026
CITI Program

Technical Skills

Programming	Python, JavaScript, TypeScript
Frameworks and libraries	React, Node.js, OpenCV
Databases	SQL, MySQL, MongoDB
Machine learning	Object detection (YOLOv5), vision-language models (BLIP, PaliGemma 2)
Security and privacy	Membership inference attacks, black-box model evaluation
Development tools	Git, GitHub Actions, Docker, UML modeling, PostHog, AWS